Vedant Shelake

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Education

D Y Patil College of Engineering, Akurdi, Pune — Aug 2023 – May 2027
B.E. Electronics & Telecommunication Engineering (Honor in AI & Data Science) — *CGPA: 9.04/10*

Experience

Machine Learning Developer (Self Projects) — Dec 2025
Personal Projects

- Designed and implemented machine learning models for prediction and classification tasks
- Performed data preprocessing, feature engineering, and model evaluation
- Integrated AI models into web applications using Flask and Streamlit
- Managed and deployed projects using GitHub, Docker, and basic AWS services

Kaggle Contributor — 2026 – Present
Machine Learning Competitions & Notebooks

- Participated in Kaggle competitions involving classification and prediction problems
- Performed data preprocessing, feature engineering, and model tuning
- Published notebooks demonstrating ML workflows and insights

Projects

AI Traffic Management System (Clear Path) | *OpenCV, YOLO, Python, Machine Learning* | GitHub — Dec 2025

- Developed an AI-based traffic monitoring system using computer vision and YOLO for real-time vehicle detection
- Processed live video streams to count and track vehicles efficiently
- Improved traffic congestion analysis for smart city applications

Fake News Detection | *Python, TensorFlow, TF-IDF, NLP* | GitHub — Jan 2026

- Developed a fake news classification model using TF-IDF feature extraction and TensorFlow
- Built an NLP pipeline including text preprocessing, tokenization, and stop-word removal
- Achieved 98.34% training accuracy and 98.04% test accuracy on labeled news data
- Evaluated model performance using accuracy metrics and real/fake news prediction

Irrigation Need Prediction System | *Python, Flask, Machine Learning* | GitHub — Apr 2026

- Built an end-to-end machine learning system to predict irrigation needs for farmers
- Performed EDA and feature engineering; Random Forest achieved highest accuracy (93.8%)
- Serialized model using pickle and deployed using Flask backend

Technical Skills

Languages: Python,C++, SQL
Machine Learning: Regression, Classification, Clustering, Model Evaluation, Feature Engineering
Deep Learning: ANN, CNN, RNN, LSTM, TensorFlow, Keras
Frameworks: Flask, Streamlit, FastAPI
Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, NLTK
Developer Tools: Git, Docker, VS Code, Jupyter Notebook, Eclipse, AWS (Basic)
Domains: NLP, Computer Vision, Predictive Modeling, Data Analysis

Certifications & Achievements

Python | Supervised ML | GenAI Buildathon (2025)
Antariksh Hackathon (2025) | Adobe Hackathon (2025)